

The Political, Economic, Institutional and Legal Dimensions of Electricity Supply Industry Reform in South Africa

Anton Eberhard

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At the Center for Environmental Science and Policy

Encina Hall East, Room 415

Stanford University

Stanford, CA 94305-6055

<http://cesp.stanford.edu/pesd>

About the Author

Anton Eberhard is a professor at the Graduate School of Business at the University of Cape Town, South Africa. His research interests include the management, reform and regulation of infrastructure industries in Africa, including the electricity, gas, telecommunications and water sectors. He has long been involved in energy policy development in South Africa and a number of other developing countries; He founded the Energy and Development Research Centre at the University of Cape Town in 1989, where he also taught until 1999. Prof. Eberhard is also a Board member of the South African National Electricity Regulator and President of the International Energy Initiative.

The Political, Economic, Institutional and Legal Dimensions of Electricity Supply Industry Reform in South Africa

*Anton Eberhard*¹

0. Introduction

The dominant trend in the evolution of the electricity supply industry (ESI) in South Africa over much of the last century was the growth and consolidation of a large and powerful state-owned, vertically-integrated monopoly. Most of the early private power producers were gradually taken over by Eskom which became responsible for new supply. The main drivers for the increased concentration and public-ownership of the industry were potential economies of scale in power plant, the requirement for large amounts of capital that could be facilitated by government guarantees, and the fact that electricity was seen to be an essential ingredient of government's industrialization strategy. At the same time, the state was also assuming a dominant role in other key infrastructure industries, including rail, air and sea transport, telecommunications, water, coal-based synthetic fuels, nuclear energy, and also the iron and steel industry. Competition and private-ownership in these sectors were thought to be non-optimal; instead the state viewed these industries as key instruments for industrialization, employment creation and economic development.

However, by the 1980s poor economic performance of state-owned enterprises (SOEs), combined with broader economic and political pressures on the apartheid state, caused government to look at reforming these institutions. The management of Eskom was not fully accountable and could plan and finance excessive generation capacity. Poor investment decisions were made. The result was massive costs to the economy and, initially, to the consumer. At the same time the vast majority of disenfranchised South Africans remained without electricity.

Consequently, Eskom's governance was overhauled and new commercial principles were embedded in the operation of the utility. Productivity was improved and the financial guarantees of government were removed. Following the democratic revolution of 1994, emphasis was given to electrification, improvements in the distribution industry, the creation of an independent regulator and the corporatization of Eskom (in parallel with reforms in other SOEs).

¹ Questions and comments may be directed to eberhard@gsb.uct.ac.za.

The reform process has been slow and modest. Eskom remains in state ownership. There appears to be no political urgency to fully unbundle Eskom and many still fear that competition and privatization will lead to higher prices and the elimination of subsidies for the poor. Eskom has played an important developmental role in bringing electricity to more people. It is run reasonably efficiently. Prices are currently low, because there has been no need for investments in new capacity for many years, and the cost of the older plant has mostly been amortized. But South Africa is living on borrowed time. Prices will have to rise to fund the next wave of new capacity, expected in 2007 and beyond.

Analysts have pointed out that there is now a reform window where the ESI can be restructured to create a more competitive and efficient environment for new investment decisions. These arguments are still not widely accepted or understood by most stakeholders. Nevertheless, the momentum for reform rolls on. An Energy Policy White Paper and subsequent Cabinet decisions have laid out a path of managed liberalization. Government has stated that it wishes to see more privatization and greater black economic empowerment. Competition is being encouraged in other sectors and is coming to electricity.

This paper/chapter explores the relationships between power sector reform and the broader political, economic, institutional and legal changes in the country. We trace the historical development of the industry and describe its key features. Next we look at the political-economy of the country and the changing environment for state-owned enterprises, including constitutional, legal, macro-economic, labor, and financial reforms. The bulk of the chapter is a section that focuses on the reforms in the ESI itself. The discussion is broken down into key, broadly chronological, episodes where the rationale for reform, the interests of the different stakeholders, the reform models and the outcomes of reform are analyzed. Finally, a concluding section highlights some of the key linkages between the reforms and the broader political-economy.

I. History of the Electricity Supply Industry in South Africa

The new electrical lights and machines that were developed in the late 19th century spread rapidly around the world and South Africa was amongst the early countries that adopted these revolutionary technologies. The first electricity lights in South Africa were installed at the railway station in Cape Town in the Cape Colony, barely two years after Thomas Edison invented the incandescent lamp in 1879. In 1882, the same year that the world's first central power station began operating in New York, the mining city of Kimberly in the Cape, installed the first electrical street-lights in the country, well ahead of London which was still using gaslights. The electricity industry expanded quickly, spurred by the capital being invested in gold mining in the Transvaal Republic in the interior.² By

² Christie, R (1984). Electricity, industry and class in South Africa. London: Macmillan.

1895, Cape Town, Johannesburg and Pretoria had steam powered and hydro electricity-generating stations.³

The first commercial central power station was built in 1897 by the Rand Central Electric Works and supplied electricity mainly to the gold-mining industry around Johannesburg. Over the next two decades many of the mines built their own power stations and some also supplied electricity to neighboring towns. In 1906 the Victoria Falls Power Company was established, but its plans to harness hydro-electric power were soon abandoned in favor of cheaper coal-fired generation.⁴ After the Union of South Africa in 1910 (combining the British colonies of the Cape and Natal, with the conquered Boer Republics of the Transvaal and the Orange Free State), the pattern of power development continued to be a mixture of municipal and private utilities, utilizing different technical standards and governed by a diversity of provincial and municipal bylaws. An example was the Transvaal Power Act of 1910 that provided for the establishment of a Power Undertakings Board with powers to license generators and distributors of electricity in a specific area.⁵

By 1920 the concept of connecting individual power stations into a single network began to be considered, as well as the electrification of the railways and adjacent towns. Government was also promoting the development of coal and iron industries and the availability of cheap and abundant electricity was seen as essential for industrialization. The Electricity Act, No 42 of 1922, created the Electricity Supply Commission. ESCOM was controlled by three (later five) Commissioners who reported to the line Minister (not parliament). ESCOM had statutory powers to establish generation and distribution undertakings to supply electricity at the lowest possible cost. It had to raise capital through the issuing of securities, debentures and stocks (although it did receive interest-bearing loans from government in the early years). ESCOM was not allowed to make a profit or a loss and was exempt from corporate income tax.⁶

The Electricity Act of 1922 also provided for the establishment of the Electricity Control Board (ECB) to regulate electricity supply undertakings. Although municipal undertakings did not require a license from the ECB, they required approval from the Provincial Administrator who, in turn, had to seek the opinion of ESCOM on whether it could not supply electricity more cheaply and efficiently. Through this mechanism, ESCOM became involved in power supply in Durban, Cape Town and many other towns. ESCOM also objected to the granting of further licenses to private producers such as the Victoria Falls Power Company and a compromise was reached whereby ESCOM would

³ National Electricity Regulator (2001). *Lighting up South Africa: a century of electricity serving humankind*. Open Hand Press, South Africa, p 20.

⁴ Typical power station sizes at this time were 40 to 60 MW

⁵ National Electricity Regulator (2001). *Lighting up South Africa: a century of electricity serving humankind*. Open Hand Press, South Africa, p 91.

⁶Ibid.

finance and own new power stations and the VFP Company would build and operate them.⁷

The general pattern of power sector development in South Africa was not very different from that in most other countries in the early decades of the past century. Large power companies integrated the full value chain from generation plants to transmission lines to retail distribution. They extinguished competition by taking over smaller companies. And, as the scale of investments and the opportunities for interconnection grew, the state became increasingly involved, progressively advancing to a dominant monopoly position in the sector.

ESCOM set about exploiting South Africa's huge deposits of inexpensive, low-grade coal. By 1930, its 100 MW Witbank station produced amongst the cheapest electricity in the world (NER, 2000). The Prime Minister of the time, General Smuts, stated that electricity in South Africa was

“As cheap as anywhere in the world, because wasteful competition had been eliminated... There will always be a very large field for private capital to operate in, but there are certain industries which experience has taught us can be driven better by Government without loss through wasteful competition”.⁸

In 1948 ESCOM purchased the largest private producer, the Victoria Falls Power Company. Apart from a few industrial and mining sector self-generators, and a few small municipal generators, ESCOM now controlled most of the power stations, as well as the high voltage transmission lines. By 1973 the transmission grid was interconnected and nationally controlled. Growth in demand was rapid. New power stations were built immediately adjacent to coal mines, mostly concentrated in the north-east of the country. The coal mines were privately owned and entered into long-term supply contracts with ESCOM. Increasingly economies of scale were sought with typical power station capacities increasing from 440 MW in the 1950s to 3600 MW in the 1980s. While efficiencies did improve, there were also unexpected costs: longer lead-times ensued as well as greater interest burdens and less flexibility in the face of uncertain demand growth.

With the oil shocks of the 1970s, the economy increasingly turned to electricity. Unprecedented growth resulted in reserve margins as low as 11% in 1975. Annual growth in peak demand between 1972 and 1982 ranged between 6 and 16%. There were also initial technical problems in the scale-up of boiler designs and the use of low-grade coal. ESCOM engineers and planners were concerned that there would be power shortages and

⁷ Steyn, G (2001). Governance, Finance and Investment: Decision making and risk in the electric power sector. DPhil, University of Sussex, p 70.s

⁸ Ibid, p 67.

they ordered even more power stations. By the end of 1983, ESCOM had 22,260 MW of generating capacity on order, double the capacity then being operated.⁹

These capacity expansions were funded through commercial debt and the issuing of bonds on the local and international markets. Government guaranteed these bonds and also provided subsidized foreign exchange cover through the Reserve Bank¹⁰. However, South Africa was experiencing a capital shortage and the cost of finance was increasing. The Electricity Act was amended in 1971 to allow ESCOM to retain substantially more earnings to build up a Capital Development Fund, subject to the approval of the State President. The eventual consequence was large price increases, disquiet amongst stakeholders and eventually the initiation of a government appointed enquiry in 1983. ESCOM's management was accused of arrogance and lack of accountability. The De Villiers Commission was critical of ESCOM's governance, its management, its electricity forecasting methods, its investment decisions and its accounting.

The Commission's recommendations led to changes in the Electricity Act in 1985 and to new Eskom and Electricity Acts in 1987. ESCOM was renamed Eskom and was reconfigured with a new two-tier governance structure, modeled broadly on the German corporate governance system. A full-time executive management board now reported to an Electricity Council comprising representatives of major electricity consumers, municipal distributors and government representatives, all appointed by the Minister. The Capital Development Fund was abolished and Eskom's old fund accounting system replaced with the standard business accounting conventions. The principle of operating at "neither a profit or a loss" was replaced by the need to "provide the system by which the electricity needs of the consumer may be satisfied in the most cost-effective manner, subject to resource constraints and the national interest."¹¹

The drafters of the new Act, who were members of ESCOM's legal department, managed to insert a clause that exempted Eskom from the requirement to have a license issued by the Electricity Control Board and thus from having its prices regulated. Effectively the Act relied on the consumer dominated Electricity Council to control prices, subject to government review and approval. However, under the influence of the strong personality of its new chairman (an influential industrialist) the Council acted more like a Board of Directors concerned with the financial health of a commercially-run company. The new management also had to re-establish Eskom's credibility and reputation with government and its customers.

⁹ Ibid, p 75.

¹⁰ Government guarantees for Eskom's International Bonds were only withdrawn in 1995. Personal communication: interview with Eskom Treasury Department, 27 January 2003.

¹¹ Eskom Act, 1987. Government Printer. Pretoria.

In an attempt to limit the extent of surplus capacity that was looming as a result of over-planning, construction of generation sets were delayed and plans for the final station were cancelled. Older plant (about 5000MW) was decommissioned or mothballed. Previous demand growth projections of 7% were scaled back. Nevertheless, maximum generating capacity still exceeded peak demand by 63% in 1992.¹² Eskom began to promote load growth through low-cost electricity contracts to energy-intensive users, including new export-oriented minerals-beneficiation investments in aluminum and ferro-chrome. However, South Africa still experiences overcapacity. No new power stations have been ordered since 1982, although the go-ahead for constructing the *last* three units of the last power station, Majuba, was delayed until 1995 and the last unit was only completed in 2001. The dates of commissioning of the major coal-fired and nuclear powered stations are shown below. Camden and Grootvlei have been mothballed, the rest are all still operating.

Table 1: Dates of commissioning of major Eskom power stations

<i>Name of power station</i>	<i>Date of commercial service First – Last unit</i>	<i>Net maximum capacity MW</i>
<i>Camden</i>	<i>1966-1969</i>	<i>1520</i>
<i>Grootvlei</i>	<i>1969-1977</i>	<i>1130</i>
<i>Hendrina</i>	<i>1970-1977</i>	<i>1900</i>
<i>Arnot</i>	<i>1971-1975</i>	<i>1980</i>
<i>Kriel</i>	<i>1976-1979</i>	<i>2850</i>
<i>Koeberg</i>	<i>1976-1985</i>	<i>1840</i>
<i>Matla</i>	<i>1979-1983</i>	<i>3450</i>
<i>Duvha</i>	<i>1980-1984</i>	<i>3450</i>
<i>Tutuka</i>	<i>1984-</i>	<i>3510</i>
<i>Lethabo</i>	<i>1985-1990</i>	<i>3558</i>
<i>Matimba</i>	<i>1987-1991</i>	<i>3690</i>
<i>Kendal</i>	<i>1988-1993</i>	<i>3840</i>
<i>Majuba</i>	<i>1992-2001</i>	<i>3843</i>

Source: Eskom Annual Reports

This pattern of overinvestment and subsequent contraction was not dissimilar to that experienced by many vertically-integrated power company monopolies during the 1970s and 1980s. When economic growth was forecast to be rapid, shortages in power supply seemed immanent and vast, new expansion projects would be undertaken, mostly within a context of investors or SOE managers assuming little risk, as the costs would be passed through to electricity consumers. But the investments were lumpy and had long lead-times. Expected growth rates were often not realized and the inevitable consequence

¹² Steyn, G (2001). Governance, Finance and Investment: Decision making and risk in the electric power sector. DPhil, University of Sussex, p 80.

was wasteful overcapacity. Planning of new plants and further investment would then stop until a new potential crisis in meeting future demand would arise. This cyclical investment pattern was clearly not optimal.

South Africa, along with many other developing countries, now faces renewed calls for capacity investment. Electricity customers have become used to cheap power from the last generation of plant expansion whose underlying capital is largely depreciated. New capacity will inevitably require higher prices and possibly more stringent environmental standards. It was inevitable that new investment models would begin to be explored.

In 1988, the first Eskom privatization study was undertaken. It was commissioned by government but managed and led by Eskom, assisted by a committee of government and industry stakeholders. The study was initiated at a time when the state was reviewing the performance of its SOEs. There was also a need to attract foreign direct investment and to counter economic sanctions and pressures against South Africa. The study indicated that privatization of Eskom was possible and suggested that it be privatized in its entirety – i.e. there were no recommendations for the introduction of competition. However, the proposals co-incided with the beginnings of the secret dialogues with the ANC about South Africa's political future, and were quietly dropped.¹³

Eskom faced a very different environment in the 1990s. The democratic revolution of 1994 unexpectedly resulted in further liberalization of the economy. The state-centered orientation of the National Party government, and also of the African National Congress during the years of the liberation struggle, gave way to a more market-oriented policy which included conservative fiscal management. The state would still play a responsible role, but this would be more transparent and predictable through improved governance and regulatory frameworks and institutions. State-owned enterprises were corporatized and subject to shareholder performance contracts and some began to be privatized. Eventually the focus had to turn once again to Eskom. At the same time, some stakeholders were becoming aware that a revolution was sweeping through the electricity industry worldwide. The old traditional model of a publicly-owned, vertically-integrated ESI was being superseded by unbundled, competitive and mostly privately-owned industry. Change was becoming inevitable.

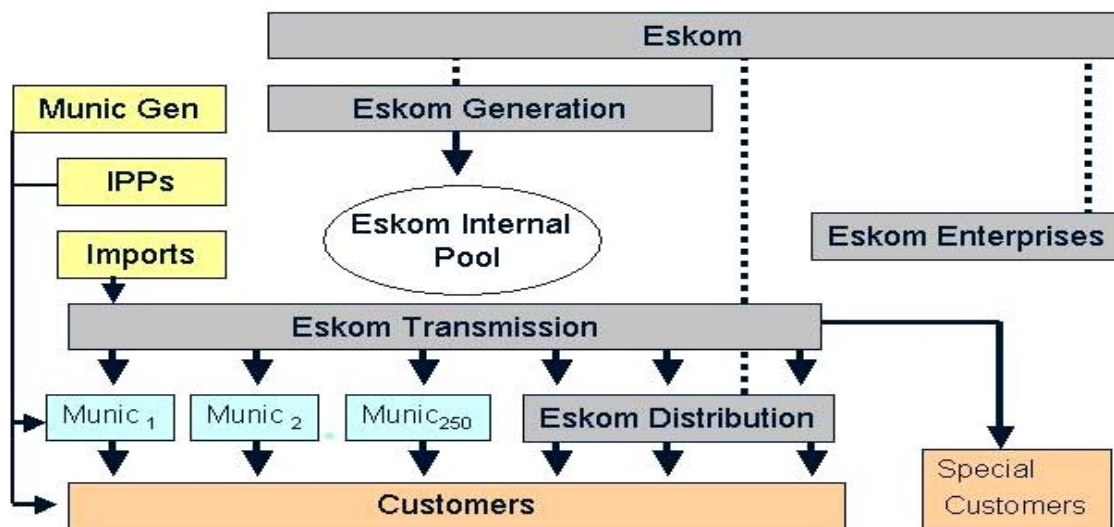
II. Overview of the Electricity Supply Industry in South Africa

The South African Electricity Supply Industry (ESI) remains dominated by the state-owned and vertically integrated utility, Eskom, which ranks seventh in the world in terms of size and electricity sales. It supplies about 96% of South Africa's electricity requirements which equals more than half of the electricity generated on the African

¹³ Personal communication: interview with the then secretary of the privatization study, 28 January 2003.

continent. Eskom owns and controls the high voltage transmission grid and it supplies about 60% of electricity directly to customers. The remainder of electricity distribution is undertaken by about 240 local authorities. They buy bulk-supplies of electricity from Eskom, with some also generating small amounts for sale in their areas of jurisdiction. A few industries have private generation facilities for their own use. 91% of electricity is generated from coal; nuclear energy accounts for 6.5%, bagasse, hydro and emergency gas turbines make up the remaining 2.5%. Total licensed operational generating capacity in 2000 was 43.1GW of which Eskom owns 39.9GW. Some capacity is mothballed and total net operating capacity amounted to 35,3 MW in year 2000. Peak demand on the system was 29.2 MW. Eskom has 24 power stations: 10 large coal fired stations dominate – most of them situated on coal mines in the north-east of the country. Nine of these stations have long-term coal contracts.¹⁴

Figure 1: Structure of the electricity supply industry in South Africa

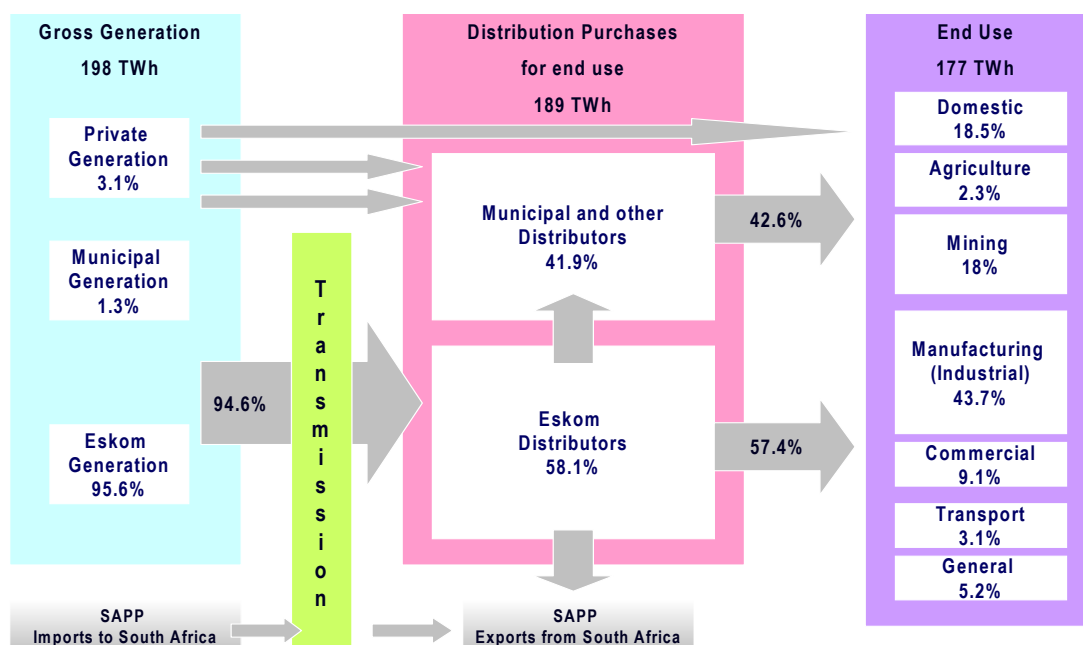


Existing industry structure

Figure2: Energy flows in the Electricity Supply Industry in South Africa 2000

Source: NER (2000).

¹⁴ The 10th large coal-fired power station, Majuba, operates at variable output on a small medium-term coal contract.



Six of these long-term coal contracts are “cost-plus” and three are “fixed price”. In the cost-plus contracts, Eskom and the coal supplier jointly provided capital for the establishment of the colliery. Eskom pays all the costs of operation of the colliery and the supplier is paid a net income by Eskom on the basis of a return on the capital invested (ROI) by the coal supplier in the colliery. The ROI is divided into two components, a fixed and a variable portion. The fixed portion is a set ROI, payable irrespective of tonnages of coal supplied and the variable portion is based upon tonnages supplied to Eskom. The ROI is generally escalated for half of the duration of the contract and is typically between 15 and 25%. In the fixed price contract, coal is supplied at a predetermined price, i.e. a base price which is escalated by means of an agreed escalation formula. There are no early termination provisions in the contracts. Generally, coal costs are regarded amongst the cheapest in the world. Although it is Eskom’s stated intention to reduce its reliance upon long-term coal supply contracts, more than 90% of Eskom’s coal is still procured in this way.

Africa’s only nuclear station is at Koeberg, 30 kilometres north of Cape Town, and is also owned and operated by Eskom. There is modest hydro capacity on the Orange River, located on two dams – and there are two pumped storage schemes which play a critical role in meeting peak demand, as well as in system balancing and control. Municipalities own 22 small power stations and back-up gas turbines, but these total only

5% of national generation capacity and generally run at low load factors. Private generators comprise the remaining 2% of capacity.¹⁵

South Africa sells electricity to neighbouring countries (Botswana, Lesotho, Mozambique, Namibia, Swaziland and Zimbabwe) representing about 3% of total net energy produced. Contractually it is bound to take electricity from Mozambique's Cahora Bassa hydro-electric station on the Zambesi. Eskom also imports some power from the Democratic Republic of Congo and from Zambia – mainly for peak load management. In 2000, Eskom imported just slightly more electricity than it exported.

Eskom operates an internal pool which generates an optimal dispatch schedule. The system operator and the procurement of auxiliary services is part of Eskom. Eskom owns the national, integrated transmission grid (with the exception of the Motraco line which is jointly owned with the utilities in Swaziland and Mozambique). The national grid comprises 27 000 km of high voltage lines, the bulk of it at 400 and 275 kV. Transmission energy losses are less than 4%. There are an additional 330 000 km of low-voltage lines owned by Eskom and local authorities. Domestic consumers account for 18% of electricity consumed; manufacturing – 44%; mining – 18% and commercial, transport and agricultural users, the rest. Eskom supplies 3.1 million customers and municipalities supply an additional 3.4 million.

Eskom makes most of its profits from the sale of electricity to its large mining and industrial customers and in bulk sales to municipalities. These three customer categories account for 82% of its revenue and 89% of its electricity sales. The average selling price to industrial customers is 1.2 US cents/kWh. In 2000, the average price to rural and residential customers was 2.8 US cents/kWh which does not cover the full cost of supply. The total annual internal cross-subsidy to these customer categories exceeds US\$ 100 million. The large municipalities, in turn, make a profit (at least US\$ 250 million per annum) from reselling Eskom electricity, which enables them to subsidise property rates and to finance other municipal services. However, many of the smaller municipalities face debt, non-payment by a substantial proportion of their low-income consumers, inefficient operations and lack of technical and managerial capacity.

Nationally, there has been an impressive electrification drive and the proportion of households with access to electricity has risen from one third, in 1993, to two-thirds in 1999. In the years 1994-2001, 3.5 million new households received electricity. About two-thirds of these connections were accomplished by Eskom and the remainder by local-authorities. The capital costs for these connections were provided by Eskom and amounted to a direct subsidy to new low-income households amounting to around US\$ 1 billion (R 9.88 billion in nominal values) over the period, with an average cost of

¹⁵ Eskom (2001). Annual Report. National Electricity Regulator (2001). Electricity Supply Statistics for South Africa. Pretoria.

connection just under US\$300 (NER, 2001). The electrification programme has resulted in significant increases in peak demand in the morning and early evenings with profound implications for future generation plant mix. The next requirement for capacity addition will be for peaking plant.¹⁶

Eskom has grown a significant R&D effort over the years. Research emerged from its engineering division and still focuses mainly on “sweating assets”, i.e. incremental improvements that solve problems, lower costs and increase efficiencies in its main generation, transmission and distribution businesses. However, Eskom has also devoted R&D resources to environmental issues, end-use technologies and alternative and future energy technologies, including the development of a new generation “pebble-bed” nuclear reactor. R&D expenditure increased rapidly in the 1990s and amounted to about US\$ 30 million in 1999, 0.8% of total revenue. The R&D capability has been transferred to Eskom Enterprises which contracts its services to Eskom Holdings and other industry participants.¹⁷

In broad terms, the electricity supply industry has supported economic development through the provision of low-cost, reliable supply of electricity. In recent years, it has also contributed to improving social equity through increasing access and subsidising supplies to the poor. However, there are a number of problems and challenges which are driving changes. These include the need to improve the performance of the distribution industry; the need to create a competitive environment for new investment in generation capacity; and to unlock the economic value in the industry. These issues will be explored further below.

III. Political and Economic Context in the 1990s

The changes in the electricity industry in South Africa over the past decade have taken place within a context of radical transformation of the political, economic and social institutions in the country. A wider range of stakeholders have been able to voice their interests more freely in the reform process. While the overall macro-economic context has shaped the direction of the reforms, some of the economic contradictions, particularly impacts on the poor, have influenced the position that some stakeholders have adopted in relation to the reforms.

After decades of institutionalised racial segregation and discrimination, the minority white government in the late 1980s faced overwhelming opposition from the majority of South Africans, and sustained international pressure, including selective economic sanctions. International lenders refused to roll-over outstanding loans. The

¹⁶ National Electricity Regulator (2001). *Lighting up South Africa*. Pretoria.

¹⁷ Hofmaenner, A (2002). *A history of energy research in South Africa*. PhD thesis. ETH, Zurich.

choices were becoming narrower. Either the leaders of the apartheid state took South Africa down a path of increasing political violence and diminished wealth, cut-off from the international community, or they could begin to negotiate a new democratic future. In 1990, the African National Congress (ANC) was unbanned and Nelson Mandela freed from prison. Soon after the first democratic elections in 1994, South Africa had a new Constitution and a Bill of Rights, widely recognized as being amongst the most progressive in the world. The ANC won the first elections with 63% of the vote, and for the first few years, there was a Government of National Unity with representatives of the other major parties in the Cabinet. In the 1999 election, the ANC increased its vote to 66%.

The effect of this shift on government policy and the role of the state was revolutionary. Politically, the apartheid government represented the interests of a small minority of the country's population, who enjoyed an extraordinary level of support in terms of job opportunities, income-levels, health and education services and general infrastructure, while the majority, mostly black South Africans, lived in poverty not dissimilar to that experienced in poor developing countries. In addition, the government in the latter decades of the apartheid-state, had become more repressive and more secretive. Much crucial policy making was undertaken in secret, and legislation was passed controlling the flow of policy-related information. By contrast the new government in 1994 represented an overwhelming majority of South Africans, and policy debate and processes were much more visible.

The ANC's economic philosophy in exile had been broadly socialist. The economic goals of the first ANC-led government were enshrined in the Reconstruction and Development Programme (RDP), an integrated policy platform which set out a Marshall Plan-like programme for social and economic advancement, centered on the development of infrastructure in poor communities. The RDP promised to redistribute land, promote affirmative action, create employment, provide houses, electricity and water, and attack poverty and deprivation.

While the RDP did deliver some important gains, for example in areas such as electricity and water provision, the new government soon faced macro-economic and capacity constraints. Under the previous Nationalist Government, the budget deficit before borrowing had soared to an unsustainable 7% of GDP. The RDP was superseded by the Growth, Employment and Redistribution (GEAR) policy, essentially a conservative, macro-economic plan that aimed to reduce the budget deficit, increase growth rates, lower inflation, reduce tariffs, stabilise the currency and create jobs. Some of GEAR's critics have labeled it as a self-imposed structural adjustment programme.

GEAR did not exactly replace the RDP, but it did signal a shift from social to economic priorities. GEAR's subsequent success has been mainly with respect to public finances and in forcing industry and agriculture to be more internationally competitive. The budget deficit has been reduced to below 2% of GDP and certain industries, such as

automobile manufacturing, have shown dramatic export growth. However, growth of the economy, although steady, is below the 6 or 7% required to deal with unemployment (currently approaching 30%, depending on which figures one accepts). Foreign and local fixed investment has been low. The currency has fluctuated and has impacted on inflation, as food prices and other commodities increasingly reflect international prices. Inflation targets of 3-6% have been difficult to achieve. The impact on the poor has been severe. Also, since 1990, more jobs have been lost than created.

The trade union alliance partners of the ANC have been particularly critical of the fiscal conservatism of the government and its policy of economic liberalization. They have argued that privatization of SOEs will impact negatively on service provision for the poor. They also fear further job losses. Government has had to factor these concerns in its reform agenda.

Government responds that its policies have avoided the economic shocks and recessions being experienced by many other emerging economies. Its debt is relatively low and the interest burden is declining, thus allowing more scope for social expenditure. The main challenges for the economy are now increasing the levels of investment, lifting growth rates, creating employment and building the capacity to increase the rate and quality of delivery of services for the poor. The state is also pushing hard to increase the ownership and participation of blacks in the economy. Currently, the value of majority-owned black companies comprises less than 5% on the Johannesburg Stock Exchange. Targets have been set in the minerals and petroleum industry of at least 25% ownership by 2010, and attention is being given to opportunities in the electricity industry.

There has been a plethora of legislation since 1994 which has sought to restructure and reform the economy and society to address the inequities and injustices of the past, and to advance principles of justice and development enshrined in the constitution. South Africa inherited a hybrid legal system, partly based on Roman-Dutch law, but also influenced by the British legal tradition. The political negotiations leading to democracy led to a new progressive Constitution, internationally admired for its protection of first generation rights (such as protection of individual liberty and property and freedom of expression), combined with 2nd and 3rd generation development-oriented rights which place obligations on the state to advance individuals' and communities' access to health, shelter, a clean environment, etc. One of the important revisions has been the rewriting of Labour Law to provide the kinds of protection afforded employees in mature social democracies.

The ANC inherited an economy with large state-owned enterprises (SOEs), particularly in energy, telecommunications, and transportation and in weapons manufacture. It retains a commitment to utilize these SOEs to fulfill national and social goals. For example, Eskom and Telkom have been tasked with the accelerated roll-out of services for the poor. The relatively good state of public finances means that National Treasury is not desperate for privatization receipts. Yet the government remains committed

to reform SOEs to make them more efficient, to create competition, to unlock economic value, to foster black economic empowerment and to attract foreign direct investment. The general direction is towards liberalizing the economy within a context of social democracy.

Within the energy sector, there is a mix of SOEs and private companies. Three-quarters of South Africa's energy derives from coal, the bulk of which is used for power generation (Eskom) and synthetic liquid-fuels (Sasol). Coal mines and distribution are privately owned. Coal prices and distribution are deregulated. The electricity industry is mostly in public ownership. The state is also involved in the nuclear industry (NECSA) and in oil exploration and synfuels from natural gas (PetroSA). Sasol and PetroSA produce about 40% of liquid fuels used in South Africa. The remainder is supplied through crude oil imports and local refining by the international oil majors. The price and distribution of liquid fuels is regulated.

The next challenge for government in the energy sector is the restructuring and liberalization of the electricity sector.

IV. Restructuring from 1990

Key challenges

Most analysts identify four broad international drivers for power sector reform internationally. First, there is the desire to improve investment and operational efficiencies that blight the performance of monopoly utilities. Second, technological innovation, particularly in efficient, less capital-intensive generation plant, and in new information and communication technologies, makes it possible to organize the industry in new ways that make it easier for new entrants to compete. Third, the need for massive new capacity expansion, places increased demands for finance that is not readily available from the public sources and which forces greater reliance on private sector involvement. And fourth, restructuring and privatization creates the opportunity for unlocking economic value and for reducing government debt. Some have identified other country-specific drivers (such as ideological commitments to liberalize the economy) or simply the felt need to follow the wave of reform that is now so powerfully sweeping through nearly all power sectors around the world.

It is probably true to say that none of these drivers are experienced strongly in South Africa. Most stakeholders believe that Eskom operates efficiently. New generation technologies such as Combined-Cycle Gas Turbines have limited application in South Africa because of its limited access to natural gas. Eskom has had no serious problem financing expansion through raising private capital (locally and internationally) through its bond issues. Public finances are well managed and National Treasury does not have a

desperate need for privatization receipts. And the impacts of international trends in power sector reform are not widely appreciated locally. Eskom proceeds very much like Electricite de France. It would prefer to stay as it is and will delay reforms as long as it is able to.

Yet, a deeper analysis of the challenges facing the power sector in South Africa, reveal issues that need addressing and which will progressively drive the reform process. Chief among these is the need to avoid the mistakes of the past when Eskom heavily over-invested in capacity expansion, and to create an industry structure that allocates risk in a manner that encourages more efficient investment efficiency.¹⁸

Investment in the electricity supply industry

At first glance, the South African Electricity Supply Industry (ESI) has performed well. Eskom supplies electricity at amongst the lowest prices in the world. The average cost of electricity generated is below 1 US cent/kWh. In recent years, it has consistently made a positive return on assets. Reliability and quality of supply are good. Average energy availability¹⁹ from its power stations has increased from 76% in 1991 to 92% in 2000, although overall generation load factors (after excess capacity management)²⁰ improved modestly from 50% to 55% over the same period. Overall thermal efficiency has remained constant over the past 10 years. Labour productivity has increased and employee numbers have dropped from over 66 000 in 1985, to 46 600 in 1991, to 32 800 in 2000. The national electricity utility is now commercially run with no recourse to the national fiscus. It raises finance through commercial debt, mostly through issuing bonds which are well supported by local and international capital markets.²¹

Eskom's recent low prices and exemplary electrification performance have left the impression that it is highly efficient and that there is no need for reform. Many would simply equate low prices with high efficiency. However, this is not necessarily the case. There may be specific factors that account for low Eskom prices compared to other international utilities and there may be little hard evidence of superior efficiency.²²

A close examination of the South African ESI shows that low prices and the ability to fund electrification have emanated from amongst other factors, very low coal prices (by

¹⁸ Business Map (2001): The electricity supply industry: economic and social effects of restructuring. Johannesburg.

¹⁹ Defined as capacity hours available x 100/total capacity hours in year

²⁰ Defined as kWh produced x 100/(average net maximum capacity x hours in year)

²¹ Eskom Annual Reports

²² Steyn, G (2001). Governance, Finance and Investment: Decision making and risk in the electric power sector. DPhil, University of Sussex.

international standards) and, until recently, exemption from taxation and dividends²³. Low prices also stem from the fact that consumers have largely amortised the debt which funded the large investment programme of the 1980s that has provided the generation capacity currently still being used. Eskom has not had to invest significantly in new generation capacity for some years and one of the largest contributors to lower overall costs (and prices) has been lower debt and financing costs. Eskom's debt to equity ratio has fallen from 2.93 in 1986 to 0.5 in 2001.²⁴ Eskom's poor investment record is revealed in the graph below:

While operational efficiencies are important, investment efficiencies often have a much more profound and long-lasting impact. Choices around fuel-type, technology, financing, investment-timing, and construction lead-times determine the primary cost structure of the generation plant. The difference that can be made by good investment decisions is often larger than the incremental productivity improvements that can be made in operational plant. The figure below shows the extent to which changes in financing charges affect overall costs and hence prices.

This analysis on Eskom's investment record is not widely shared in South Africa and the predominant view of most stakeholders now is that Eskom is efficient and that generation prices are amongst the lowest in the world. Few recall the debacle of Eskom in the 1980s, the high price hikes, and the criticisms of Eskom's governance and management. Few understand the consequences of the massive over-investment. After years of price increases, Eskom was able to reduce prices in the early 1990s – as it had little need for further investment and progressively faced lower finance charges. Eskom entered into a pricing compact with government between 1991 and 1995 and reduced prices by 25% in real terms. The overall standing and image of Eskom in the 1990s is much improved.

²³ Steyn, G (2000). A competitive electricity market for South Africa: the need for change and a strategy for restructuring South Africa's electricity supply industry. Paper prepared for the Department of Minerals and Energy. February. Pretoria

²⁴ Eskom Annual Reports

Figure 3: Historical growth in maximum demand and capacity at Eskom²⁵

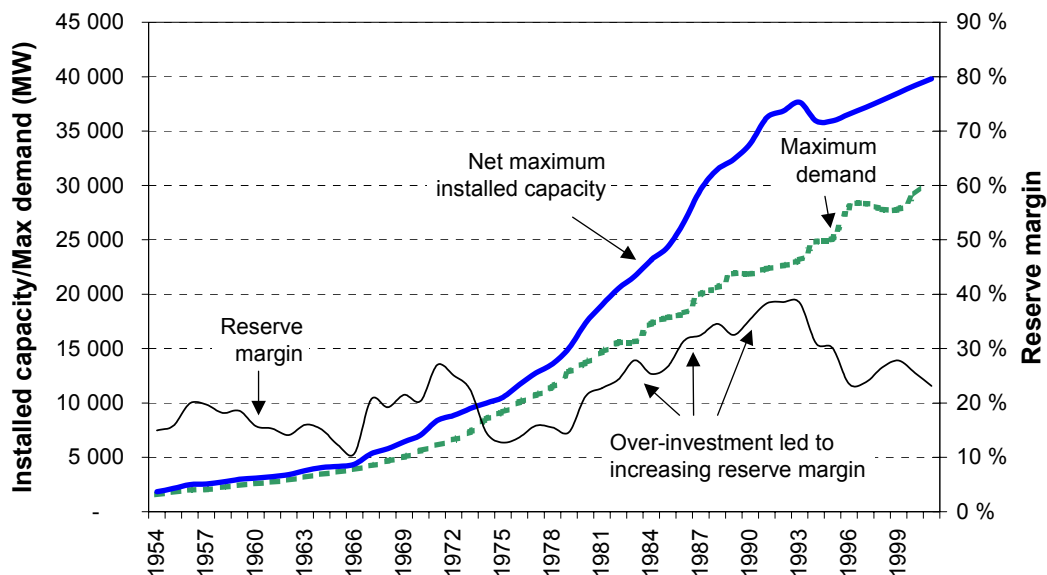
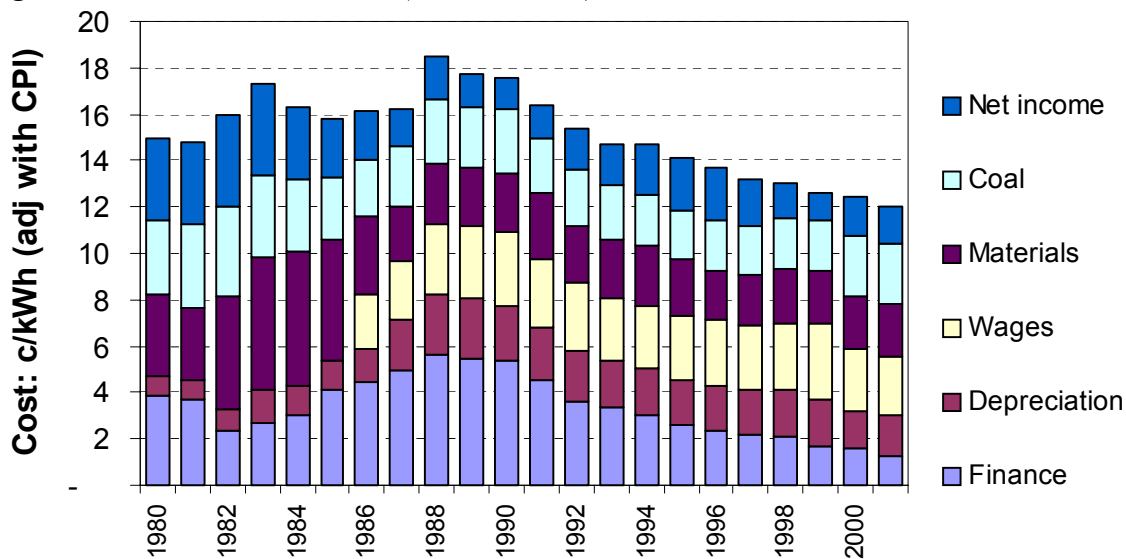


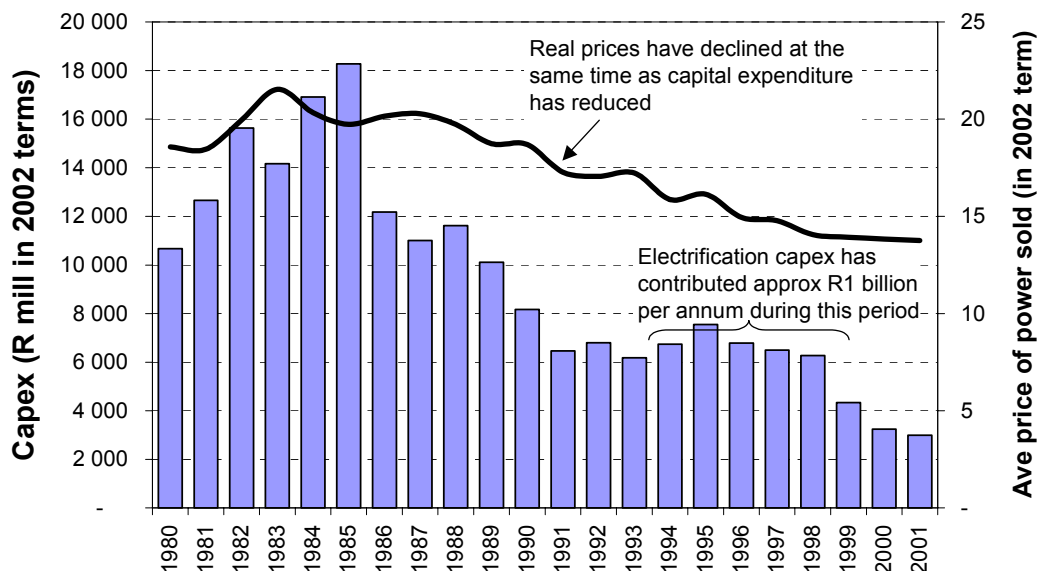
Figure 4: Cost trends at Eskom (SA cents/kWh)²⁶



²⁵ Econ (2002). Benefits and risks of power sector reform. Commissioned by the Department of Public Enterprises, South Africa. Memo 29/02. Oslo.

²⁶ Ibid.

Figure 5: Eskom capital expenditure and price trends²⁷



However, Eskom is now keen to see prices rise to levels that can support the new investment that is now necessary. Analysts (and a small minority of officials in government) have begun to understand that current prices are economically unsustainable.²⁸

Eskom's Integrated Strategic Electricity Plan suggests that by 2025 total maximum demand could rise to between 40 and 70 GW depending on electricity growth scenarios. A moderate growth scenario would imply a total system maximum demand of 52GW, almost double the current maximum, implying new investment of at least US\$6 billion and probably more, depending on technology choices. New capacity might be needed as soon as 2007, although other scenarios define 2011 as being critical. Important investment decisions will have to be made in the next few years. The primary policy challenge is to design an industry structure that provides the incentives to optimise investment efficiencies in the future.²⁹

Inefficiencies in the distribution industry

There has been a growing consensus amongst stakeholders that problems in the electricity distribution industry also need urgent attention. There are simply too many

²⁷ Ibid.

²⁸ Econ (2002). Electricity Price Scenarios for South Africa. A report to the Department of Minerals and Energy, South Africa. Oslo.

²⁹ National Energy Regulator (2002). National Integrated Resource Plan. Pretoria.

small, non-viable, poorly-run municipal distributors. Many problems stemmed from the creation of separate black-local authorities in the apartheid era. Most were not financially viable, and their electricity departments struggled with lack of capacity, few income-generating industrial customers, and a huge back-log in new connections for low-income consumers. Even after the amalgamation of local authorities, many still face severe debt problems, including non-payment from customers. Some have had problems paying Eskom. Investment in maintaining networks is inadequate and the security and reliability of supply is being compromised. Individual distributors are unable to meet the financial demands of the electrification programme. Tariffs, for the same customer categories, vary widely between distributors. It is difficult to attract skilled, motivated and adequately paid employees and managers in the industry. And it is difficult for the regulator to regulate more than 240 distribution entities effectively. Rationalization is an imperative as well as ensuring that electricity is accessible and affordable for the poor. At the beginning of the 1990s this was the issue of overriding concern.^{30 31}

No champion for reform, but pressures build to accelerate black economic empowerment

There has never been a single, powerful champion for reform, neither in government, nor amongst the stakeholders. There has also been much institutional upheaval. Key government officials in the Department of Minerals and Energy and in the Department of Public Enterprises have seldom lasted more than 2 or 3 years in their positions. Institutional memory and capacity has often been lost and has then had to be rebuilt. As a consequence, the reform process has been slow, convoluted and frustrating. Perhaps the one driver for reform that is clearly articulated in the political domain is the need to accelerate black economic empowerment. Eskom assets are seen as attractive and a portion could be offered on a preferential basis to black South Africans. If new private players are introduced into the industry, then it makes sense to put in place a competitive market framework.³²

Despite the slow progress, some key milestones have been reached. The 1990s saw the launch of a major electrification programme with structured subsidies, a new National Electricity Regulator was established to protect the interests of consumers and to promote efficiencies in the ESI. A new national energy policy was finalised, including broad policy objectives and restructuring principles for the electricity sector. A decision was reached on rationalising the distribution industry. Eskom has been corporatised and a broad industry restructuring plan agreed to. And finally, detailed work has begun on the design and implementation of a competitive electricity market.

³⁰ Media briefing by Minister of Minerals and Energy, Phumzile Mlambo-Ngcuka, 31 July 2001, Pretoria.

³¹ Department of Minerals and Energy (2001). Electricity Distribution Industry Restructuring Blueprint Report. Pretoria.

³² Department of Minerals and Energy / Department of Public Enterprises (2000). Options for reform of the electricity supply industry in South Africa. Internal Government Paper. Version 3.0, July, Pretoria.

The next section highlights the key milestones in the reform of the electricity industry, the motivations and rationalisation behind each reform episode, and the competing stakeholder interests which shaped the reforms.

Key reform milestones

1993/4 – Serving the poor: an accelerated national electrification programme

The first significant change in the 1990s in the electricity industry was the recognition that urgent attention had to be given to providing electricity to the majority of South Africans, and that a plan and programme to achieve this had to be put in place. With the exception of some studies³³ in the 1980s that highlighted the inequity of electricity provision, little data existed documenting the demand from un-served households. Nearly all white South Africans, including remote farms, had electricity connections; few black households had access. Some researchers³⁴ began to map out what a national electrification programme might look like and argued that it would be important to restructure the inefficient distribution industry. The changes in the political landscape in South Africa, after 1990, lent some urgency to these calls for action. Energy policy had to be re-directed from the apartheid focus on energy security, to a new focus on improving social equity through widened access to affordable services. The South African Energy Policy Research and Training Project was initiated in 1992 at the Energy and Development Research Centre (EDRC) at the University of Cape Town – and played a major role in defining this new energy policy paradigm.³⁵

Eskom, in anticipation of the shift to political democracy, and sitting with excess electricity generating capacity announced, in 1991, the target of electrifying 700,000 new households by 1997.³⁶ The programme was backed by a call from its CEO, Dr Ian McCrae, for “electricity for all”. There were some high-profile initiatives in Elands kraal, Orange Farm and Soweto, which in hindsight can be seen as an attempt to position Eskom favourably in relation to the ANC and a possible new democratic government. But overall progress was slow. EDRC called for an accelerated programme of electrification which

³³ E.g. Eberhard (1984). Energy and poverty in urban and peri-urban areas around Cape Town. Second Carnegie inquiry into poverty and development in Southern Africa. Conference Paper No 155, University of Cape Town. When Eberhard asked municipal electrical engineers for maps and plans of areas that had access to electricity and those that did not, they were unable to produce any coherent or integrated picture. Planning for those who were unserved was simply non-existent.

³⁴ E.g. Dingley, C (1990). Electricity for All. Monograph. Department of Electrical Engineering, University of Cape Town. Theron, P, Eberhard, A & Dingley, C (1992). Electricity provision in the urban areas of South Africa: towards a new framework. *Urban Forum* Vol 2, No 2.

³⁵ Eberhard, A & van Horen (1995). Poverty and Power: Energy and the South African State. London: Pluto Press

³⁶ Eskom (2001). Annual Report. Johannesburg.

would peak at 500,000 connections per annum and would electrify 85-90% of South Africans by 2010. The proposals were supported by detailed modeling and included recommendations for financing and institutional change.³⁷

In February 1992, EDRC convened a national meeting on electrification on behalf of the un-banned African National Congress.³⁸ The seminar brought together members of the industry with political parties, trade unions and civic organizations. From that meeting came the idea for a national conference on electrification and the creation of a negotiating forum involving all stakeholders. After two national conferences, involving more than 70 organizations, the National Electrification Forum (NELF) was launched in May 1993. The Management Committee of NELF consisted of representatives of the Association of Municipal Undertakings, the African National Congress, the Chamber of Mines, the Department of Mineral and Energy Affairs, the Development Bank of Southern Africa, Eskom, the National Union of Metal Workers, the National Union of Mine Workers, the South African Agricultural Union, the South African National Civic Organisation, and the United Municipal Executive.³⁹ NELF established working groups and initiated a number of studies, including the National Electrification Economic Study⁴⁰, which further developed a range of scenarios and assessed their economic impact. All stakeholders supported an accelerated electrification programme.

The ANC's Reconstruction and Development Programme, partly influenced by the above work, and partly influenced by ANC activists in EDRC, formalized the goal of electrifying 2.5 million new homes between 1994 and 1999, a goal that was exceeded by the new democratically elected government.⁴¹

Table 2: Number of new connections to low-income households since 1991⁴²

Year	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	TOTAL
Eskom	31035	145522	208801	254383	313179	307047	274345	280977	293006	250801	206103	2565199
LA's	51435	74335	107034	164535	150454	137534	213768	136074	144043	139780	127255	1446247
Farmworkers		12698	16074	16838	15134	9414	11198	10375	6241	6438	3560	107970

³⁷ Eberhard, A & van Horen (1995). *Poverty and Power: Energy and the South African State*. London: Pluto Press

³⁸ Theron, P (ed) (1992). *Proceedings of the ANC National Conference on Electrification*. Elan Press, Cape Town.

³⁹ National Electricity Regulator (2001). *Lighting up South Africa: a century of electricity serving humankind*. Open Hand Press, South Africa, p 70.

⁴⁰ National Electrification Economic Study (1993). *Financing requirements of national electrification scenarios*. Finance and Tariffs Working Group. National Electrification Forum. Johannesburg.

⁴¹ African National Congress (1994). *Reconstruction and development programme*. Johannesburg: Umanyano Publications.

⁴² National Electricity Regulator (2001). *Lighting up South Africa: a century of electricity serving humankind*. Open Hand Press, South Africa.

National Electricity Regulator (2001). *Lighting up South Africa. Electrification Statistics*. Johannesburg.

Total	82470	232555	331909	435756	478767	453995	499311	427426	443290	397019	336918	4119416
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Up until year 2000, the entire electrification programme was funded by Eskom, either through internal subsidies – or through transfers to an electrification fund that the National Electricity Regulator allocated to municipalities. The average annual capital expenditure on this programme has been around US\$ 110 million and the average cost per connection has been around US\$ 290. The electricity industry has been able to fund and cross-subsidize this massive electrification programme, largely because there is a substantial industrial customer base that accounts for the bulk of electricity sales. Unit cross-subsidies from these customers have been proportionately small and politically acceptable.

National policy has been that the capital cost of connections should be subsidized. However, there have also been unplanned cross-subsidies in operating costs. At the beginning of the programme it was estimated that the average monthly consumption of newly connected, low-income households would be 350 kilowatt hours per month (compared with an average of 750 kilowatt hours per month for a middle-income family in South Africa). In practice, average monthly consumption has been less than 100 kilowatt hours and often as low as 50 kilowatt hours per month. Government has now decided to grant 50 kWh per month free to poor consumers who accept load-limited supplies. The grant is made from National Treasury and is allocated to local government.

Nearly all of these new connections have used pre-payment technology – customers buy tokens or top-up electronic cards to activate their electricity dispenser. The costs of the electricity supply and use were to be recovered through a flat energy unit charge. Many connections involve informal houses (shacks) and use pre-wired “ready boards” – typically with a few lights and plug points.

As government begins to reform the power sector, it has moved to secure the national electrification programme through establishing a separate National Electrification Fund in the Department of Minerals and Energy. Eskom was corporatized in 2001 and now pays taxes and dividends. Although the taxes paid by Eskom in 2001 did not cover the amount allocated to the Electrification Fund by Treasury, it is expected that taxes and dividends will exceed electrification contributions in the future. A National Electrification Programme Management Unit has been established as well as a National Electrification Business Planning and Operations Management Unit. Electrification targets will continue to be set for the industry.

This experience is important as it demonstrates that the meeting of social goals and public benefits is independent of industry structure. Electrification was carried out by the old vertically-integrated, publicly-owned utility, Eskom, and by local government distributors. The electrification programme will continue, despite the pending liberalization of the electricity market in South Africa. Explicit policy and regulatory instruments have

been put in place to ensure the continued commitment to move to universal access to electricity in South Africa.

In addition to the grid-electrification programme, there has been an active off-grid programme using photovoltaic technology. Between 1994 and 2000, 1,350 schools were electrified with off-grid systems. Many rural health clinics have been equipped with solar systems. In addition, government has awarded subsidy concessions to private industry service providers in five geographic areas to supply solar home systems as well as supplementary fuels such as liquid petroleum gas. These are not geographically exclusive concessions; other companies may also operate in the areas. However, the concessionaire in each geographic area will receive a subsidy of US\$ 320 per installation. The rationale is to assist service providers in building up adequate service infrastructure and to move towards financial sustainability. Supply targets and service standards have been set and performance will be monitored.

The concession contractual framework has been less than perfect. For example, there was little entry competition, and firms were not required to bid competitively on subsidy requirements. The opportunity to encourage efficiency and lower costs has not been maximized. Nevertheless, considerable innovation is emerging in the systems and vending technology employed. Most suppliers have adopted a fee-for-service approach rather than the outright sale of solar home systems.

The electrification programme in South Africa demonstrates that it is possible to make substantial progress in widening access to electricity services for the poor, even as electricity industries are restructured. The electrification programme was driven by the advent of democracy and a political commitment to provide services for the poor. It was made possible by an electricity industry that was technically competent and financially strong. And it has been put on a sustainable basis through explicit policy and regulatory instruments that will give expression to government's social goals, even when the electricity industry is unbundled and possibly privatized.

1995 – A new electricity regulator

The next crucial reform milestone was the clarification of government's role in relation to the ESI through the establishment of an independent regulator. No longer would prices be approved by politicians. The regulator would operate within a clear and transparent legal mandate to license all electricity suppliers, to approve their tariffs, monitor the quality of supply and settle disputes. Interestingly, the proposal to establish a regulator came during a period of uncertain political transition. The proposal came initially not from government, but from the stakeholders themselves. .

As previously mentioned, regulation of the electricity industry dates back to at least 1910. The first national legislation in the Union of South Africa was the Electricity Act of

1922 which set up the Electricity Control Board (ECB). The ECB's powers were constrained by the fact that they did not have direct control over municipal electricity undertakings. The Electricity Act of 1987 also exempted Eskom from having to obtain a license. In the 1990s, the regulatory framework was substantially revised.

One of the key recommendations to emerge from the National Electrification Forum was that the ECB should be replaced by a National Electricity Regulator (NER) with wider powers to regulate the electricity supply industry. Some of the stakeholders, such as Cosatu, hoped that the regulator would be able to implement NELF recommendations such as the proposals to rationalize the distribution industry. NELF recommended that the NER should report to parliament via the Minister of Minerals and Energy who would also approve its budget. The budget would be funded from a levy on electricity sales from generators.

In October 1994, the Cabinet approved the NELF recommendations for the establishment of the NER and the Electricity Act was amended in November 1994 and again in 1995 when the NER was constituted legally as an independent institution. The Act removed the license exemptions previously enjoyed by state departments, local authorities and Eskom. The only significant exemption that remains is for persons (excluding local authorities) selling less than 5 gigawatt hours of electricity per annum and self-generators with capacities of 500 KW or less.

The NELF management committee was given the opportunity to nominate and interview candidates to the NER Board which was to comprise 7 to 9 part-time members. The interviewing panel's recommendations were accepted by the Minister. The chair of the NELF management committee (who was also previously Eskom's CEO) became the first Chairman and Chief Executive of the NER.

In hindsight it is rather curious that the one concrete accomplishment of NELF was the establishment of an independent regulator with jurisdiction over the entire electricity supply industry. The debate on power sector reform was not very well developed. The focus was on distribution and on electrification. Yet there were some constituencies within NELF and its working groups, mainly researchers/analysts and some senior Eskom staff, who recognized that it was an anomaly for Eskom, with its virtual monopoly in generation and transmission, and for local authorities, with their distribution area monopolies, to be exempt from regulation. Some had traveled and looked at ESI reform in other countries and one had attended the first NARUC regulator training course. Perhaps there was the desire to create a more predictable and transparent operating environment in a period of political and institutional uncertainty.

The Electricity Act empowers the NER to:

- issue or revoke licenses to generators, transmitters and distributors; and to determine whether licensees are competent to supply
- regulate tariff levels and structures
- regulate quality of supply and service standards
- collect information
- settle disputes between suppliers and between suppliers and customers
- approve or reject applications for expropriation of property for facilitation of electricity supply
- advise the Minister on any policy matter

Many of the initial staff in the NER were ex-Eskom employees. There were significant management and staff changes in 1997 and 1998 when the majority of these initial employees left. A new Board was appointed in 1998 and management and corporate governance failures soon became apparent. After considerable conflict, the entire new management team was sacked in late 1998, and from 1999 the challenge was to build a new institution equipped with the requisite skills, values and commitment. A third Board was appointed in 2001 and the NER has emerged as one of the more respected independent regulatory institutions in the African continent. Nevertheless the NER still faces huge challenges in terms of building sufficient capacity to effectively regulate Eskom and the many municipal distributors, as well as preparing for a new, competitive market in the future. Creating new, stable and competent institutions in developing countries and emerging economies is a formidable task, particularly when there is little tradition of independent regulation.

The role of the regulator will change as the ESI is restructured to introduce competition. South Africa has established a broad competition (anti-trust) policy and legislative framework which is captured in the Competitions Act (1998).

A memorandum of understanding was concluded between the NER and the Competition Commission in 2002. It makes provision for the establishment of a joint working group that will demarcate more clearly their respective jurisdictions and functions. The NER will continue to regulate the natural monopoly components of the industry, viz. the transmission and distribution wires. It is also likely to assume responsibility for market surveillance and then to work with the Competitions Commission around any problems or challenges concerning market power and abuse.⁴³

⁴³ Eberhard, A A (2000): Competition and regulation in the electricity supply industry in South Africa. Competition Commission, Pretoria.

The NER does not currently have any direct jurisdiction over environment or health and safety issues. Overall regulatory responsibility for the environmental governance of Eskom and other electricity suppliers vests with the Department of Environmental Affairs and Tourism in terms of the National Environmental Management Act, 1998 and the Environment Conservation Act, 1989 and its Regulations. The first act defines the environment as a functional area of concurrent national and provincial legislative competence. It further provides that environmental management must place people and their needs at the forefront of its concern. Also entrenched in section 2(4) is the principle of sustainable development, which requires the integration of social, economic and environmental factors in the planning and implementation of decisions to ensure that development serves present and future generations.

The regulations (R1183) associated with the National Environmental Management Act, 1989 specify that new generation plants require authorization from a competent authority and lays out in detail the steps that have to be followed. They have to engage independent consultants to undertake Environmental Impact Assessments (EIAs) which require the approval of the relevant local authority. The Department's officials work closely with the applicant and its consultants on the final EIA that goes for approval. The record of decision, which eventually authorizes the applicant to undertake the proposed activity, also contains conditions, much like the NER's license conditions. An obligatory condition always contained in the record of decision is that the applicant must prepare an Environmental Management Plan. Applicants that are refused may appeal to the Minister.

Some subsequent monitoring does take place. This regulation is done by the Chief Air Pollution Control Officers (CAPCO) from the Department's nine provincial offices, who work in close co-operation with the water pollution control officers from the Department of Water Affairs and Forestry. CAPCO issues registration certificates for individual power stations. These certificates state the actual quantity of particulate emissions that may be emitted from, for example, a power station, during a 31 day period.

Officials from the Department of Water Affairs and Forestry are also apart of the EIA procedure referred to above. Their role is to ensure that all the requirements of the relevant water legislation are met. The most significant regulatory role they play is the issuing of licenses for each water use, in terms of the National Water Act No. 36 of 1998 and its regulations. So, for example, in the case of a hydro-electric plant, a permit for the scheme must be obtained from this department.

A new Bill is envisaged for a combined energy regulator and this bill may require the NER to play a stronger role in environmental and health and safety regulation.

1998 – A paradigm shift in energy and electricity policy

The foundation for the reform of the energy sector as a whole was laid down in the Cabinet approved White Paper on Energy Policy released in December 1998. This new policy framework represented a complete break from the past. It presented a new paradigm for the development of the energy sector.

Two main threads informed energy policy through most of the past century. From the 1920s, energy was seen as an input that could be provided at very low cost by the state, which would facilitate the development of mining and primary industry. The second thread, which became increasingly significant during the 1970s and 1980s, was that of energy security. A white-led minority government, facing the opprobrium of the world and a United National-led oil embargo, spent many billions on the Sasol and Moss gas synthetic petroleum fuel plants and on building up a local nuclear capability through the Atomic Energy Corporation (now the Nuclear Energy Corporation of South Africa, NECSA). Both programs were begun as part of national development and modernization strategy – however, both became icons of the attempt of the apartheid government to secure energy independence. South Africa built important components of the nuclear fuel chain, including uranium conversion and enrichment and fuel-rod fabrication. It also constructed atomic weapons. Yet, South Africa never became fully self-sufficient in either petroleum or nuclear fuels and, in spite of international sanctions, was able to import these fuels more cheaply than it could produce locally. A third effort at enhancing energy security was the accelerated programme by Eskom for building large power stations using low-grade coal. We have already pointed to the enormous cost of this over-building. Energy policy under apartheid was a costly failure.

The policy process during this period was characterized by excessive secrecy which made rational and public debate on energy policy impossible. The decisions to build Sasol II and III, Moss gas, and the uranium enrichment plants, were all made in secret and announced as a *fait accompli*, despite the reservations of those concerned about the financial implications. The Petroleum Products Act of 1977 provided for jail sentence of ten years for making any comment or disclosing the source, transportation, storage or consumption of any petroleum produce in South Africa! Similar secrecy prevailed in the nuclear industry, and Eskom's power stations were designated restricted security zones. Energy policy in the apartheid years ignored democratic process, good governance, financial sense and the needs of those who most lacked adequate energy supplies.

The shift to a new energy policy was supported by three developments, namely the intellectual development of a paradigm, the political process of legitimizing the new paradigm, and its structural realization in the economy and the society.

The intellectual origins of this new paradigm can be traced to the 1980s with a number of researchers who had developed an interest in the energy problems of low-

income households.⁴⁴ A pool of expertise was built up in the Energy for Development Research Centre at the University of Cape Town, which launched a large energy policy project in anticipation of a new government.⁴⁵ A unifying theme of this work was the emphasis on the “three e’s, namely economic efficiency, social equity and environmental sustainability.

The political aspect, crucial to the adoption of the new intellectual orientation, began with the un-banning of the ANC in 1990. A productive relationship between researchers (many who were ANC members) and the ANC policy desk led to the framing of the energy component of the RDP and the early policy work in the electricity, petroleum sectors and nuclear sectors. After the 1994 elections, under the conditions of the formation of the Government of National Unity, the portfolio of Minerals and Energy was awarded to the incumbent National Party. The new minister was resistant to change, leading the ANC into the rather paradoxical position of opposing ministerial policy (or the lack of policy) through a parliamentary portfolio committee which it controlled. Energy policy, at this time, was thus driven not by the Department or Ministry but by Parliament, supported by energy analysts and researchers. Effectively the Department was cajoled into commissioning researchers to write a green paper (discussion document) on energy policy and then to assist with the development of a White Paper. After a consultation period, which included a National Conference of all stakeholders, public hearings in Parliament, and after the National Party left the government, a new White Paper on Energy Policy was published in December 1998.⁴⁶

In terms of structural change, the shift has been much more problematic and slow. The only significant shift in resource allocation had been the funding allocated to the national electrification programme. The structure and resource allocation of the main state-owned energy industries – electricity, nuclear and parts of petroleum – had remained largely intact. The main challenges have been to match policy with capacity, with resources, and with real structural transformation. Yet the White Paper has established an inexorable momentum. It has remained the touchstone of energy policy and has set the direction for restructuring and reform. In the White Paper⁴⁷, the electricity sector’s objectives are stated as:

- “Improving social equity by specifically addressing the energy requirements of the poor

⁴⁴ Hofmaenner, A (2000). An anthology research: the Energy and Development Research Centre 1989-1999. Cape Town: Elan Press.

⁴⁵ Eberhard, A & van Horen, C (1995). Poverty and Power: Energy and the South African State. London: Pluto Press.

⁴⁶ Marquard, A & Eberhard, A A. Towards energy equity, efficiency and environmental sustainability in South Africa: policy challenges. Energy for Sustainable Development, Vol IV, No 4, 3-7, 2000.

⁴⁷ Department of Minerals and Energy (1998). White Paper on the Energy Policy of the Republic of South Africa. Government Printer, Pretoria.

- Enhancing the efficiency and competitiveness of the South African economy by providing low-cost and high quality electricity inputs to industrial, mining and other sectors
- Achieving environmental sustainability in both the short and the long-term usage of natural resources.”

The White Paper also emphasises objectives of improving energy sector governance and achieving energy security through diversity of supply. It is further stated that in order to ensure the successes of the electricity supply industry as a whole, various developments will have to be considered by government over time, namely:

- “Giving customers the right to choose their electricity supplier
- Introducing competition into the industry especially the generation sector
- Permitting open non-discriminatory access to the transmission system, and
- Encouraging private sector participation in the industry”

The White Paper also states that Government believes that Eskom will have to be restructured into separate generation and transmission companies and that Government intends separating power stations into a number of companies. Distribution will be rationalised by combining Eskom and municipal distributors into a limited number of Regional Electricity Distributors. The White Paper also affirms the importance of independent regulation.

Thus the model of power sector reform laid out in the White Paper mirrors the standard or ideal model being followed internationally: vertical and horizontal unbundling in order to separate out the potentially competitive components of the industry (generation and retail supply) from the natural monopoly components (transmission and distribution wires); the introduction of competition through new private players; non-discriminatory, open-access to transmission; and independent regulation.

The White Paper was broadly supported by organised business, by energy-intensive users, and, surprisingly, by Eskom, who perhaps was not quite used to engaging with government and in policy processes in the public eye. Eskom has stated that it supports the introduction of competition, but it consistently resists any proposals that it should divest more than 30% of its generation stations. At times it seems to conflate the issue of competition with privatisation and has argued that the introduction of strategic equity partner in Eskom should be considered. It has also tried consistently to delay the restructuring process. It has also resisted the timing of separating out Transmission. It argues that placing Transmission into a subsidiary company within the Eskom group

would be a sufficient degree of unbundling. At times it has also presented alternative models for distribution, giving Eskom a more prominent role in the restructuring.

Major opposition to the proposals in the White Paper were presented to Parliament by the Congress of South African Trade Unions (Cosatu). In essence, they opposed privatisation and argued that Eskom should remain a vertically-integrated, publicly-owned utility and should be used as an agent of government to provide low-cost electricity services to all, especially the poor. They supported rationalisation of the distribution industry, but into a single national distributor.⁴⁸

In many respects, the sentiments in the White Paper were ahead of their time. The analysis and original formulation had come from the research community working with officials in the Department of Minerals and Energy. There was no broad-based understanding of the challenges facing the generation sector or the need for reform. The stakeholders had had an opportunity to present their views. In some minor instances, these were incorporated. But in many cases, the process for finalising the policy was opaque – it was intellectually-led, based on a belief of what needed to happen in the future. The importance of the White Paper is that it created a reference point for agreeing on principles of reform. It also mapped out the broad direction and created a momentum that has continued to the present.

Ongoing - rationalization of the electricity distribution industry to create viable distribution companies

One of the key concerns of most of the stakeholders represented in NELF, was the restructuring and rationalization of the electricity distribution industry (EDI) to improve efficiencies, lower costs and to position it to meet the ambitious tasks of the national electrification programme. These concerns were not always shared by local government who feared the loss of surplus income from the sale of electricity. Local government, as represented by the South African Local Government Association and through the Association of Municipal Electrical Undertakings has consistently been ambivalent in their support for the need for rationalization. Eskom has at times also resisted the possibility of losing its distribution functions. The unions, on the other hand, are strong proponents for establishing one single, publicly-owned national distributor.

After the elections of 1994, many of the negotiating forums that had been set up in the transition period to democracy were dissolved. NELF was disbanded in February 1995. In the meantime, the National Electricity Regulator has been established and was a potential vehicle for furthering reform. The Board of the regulator considered whether the rationalization of the distribution industry could be forced through the licensing process or

⁴⁸ Tinto, E (2002). Restructuring South Africa's Electricity Supply Industry. MPhil. University of Cape Town.

whether government intervention was required. It decided on the latter option and requested permission to convene an Electricity Working Group (EWG) to further develop proposals to restructure the electricity distribution industry. EWG comprised representatives from the NER, government, Eskom and the municipalities, but excluded unions and civic organizations. They evaluated the work of NELF and submitted a report to government with specific options for restructuring the industry. Government then set up an internal Electricity Restructuring Interdepartmental Committee (ERIC) which made recommendations to the Cabinet. After a long and convoluted process, the Cabinet approved in principle, in May 1997, and in June 1999, the consolidation of the EDI into the maximum number of financially viable and independent Regional Electricity Distributors (REDs).

After another delay, Government, in early 2000, appointed a consortium led by consultants PriceWaterhouseCoopers to undertake detailed modeling to determine the number and boundaries of the REDs and also the detailed planning to undertake the rationalization. Six working papers were drafted: definition of the REDs boundaries; ownership, governance and legal status; asset valuation and transfer; regulation; tariffs and financial transactions; and organization and human resource issues. Workshops were held with stakeholders to canvass their response to the proposals. By June 2000, PwC had produced a seventh working paper: Consolidated Emerging Views. The work of the consultants was overseen by the Electricity Distribution Industry Restructuring Committee (EDRIC), comprising relevant government departments, Eskom, local government and the NER. EDRIC built on the PwC report to produce a “Blueprint for EDI Reform”.⁴⁹

In June 1999, the Cabinet approved the EDI Holding Company model as the transitional mechanism that would take the EDI from its current state to an end-state of six viable and independent REDs. In November 2000, the EDI restructuring proposals were presented to the joint meeting of the Cabinet Committee for the Economic and Social Sectors, where fairly fundamental questions and concerns were raised. It was clear that all relevant Ministers had not been properly briefed, and some had not been part of the process at all. There was no appreciation of the long stakeholder process that had led to the proposals. However, the report had not highlighted some of the key political concerns such as impacts on local government finance and the securing of the national electrification programme.

Interestingly, the Cabinet directed that a consolidated proposal addressing the reform of not just the distribution sector, but the entire supply industry also be prepared. In order to deal with these concerns the Minister of Minerals and Energy appointed a Reference Group in January 2001 to review the revised EDI restructuring proposals and address issues raised by the Cabinet. The EDI Blueprint report was revised to address the questions and concerns of the Cabinet and to incorporate comments from the Reference

⁴⁹ Department of Minerals and Energy (2001). Electricity Distribution Industry Blueprint Report. Pretoria.

Group. Finally in May 2001, the Cabinet considered the consolidated proposals on electricity sector reform and agreed that local government and Eskom distributors should be rationalized into six Regional Electricity Distributors (REDs) and that an EDI Holdings Company would be established to manage the transition. It was still concerned about the possible impact on local government finances and tasked a team to investigate this. It was also concerned that all stakeholders should be adequately consulted on the implementation of the restructuring, including the NER, Eskom, organized labor, customers, provincial and local government.

The Minister of Minerals and Energy managed to get the NER, Eskom and the SALGA to sign a memorandum of understanding agreeing to the implementation of the restructuring (in 2002), but elements with local government have remained ambivalent and even hostile to the proposal. Some have argued that the Constitution and the Municipal Systems Act (2000) entrench their rights to electricity distribution. There have been threats to take the matter to the Constitutional Court. Different elements of the ruling African National Congress have expressed different interests. While the leadership asserts the importance of a national solution to the problems of electricity distribution, those involved at local government fear losing their influence. As a result there has been no dynamic leadership of the reform process and steps towards restructuring have been slow. Discussion on rationalizing the distribution industry has been taking place for a decade. It has been an extraordinarily long process. Often new leadership has joined the debates without the benefit of previously reached understandings and agreements. There were many lost opportunities to drive the process to a political conclusion. Even after the latest definitive Cabinet decisions, it has taken more than a year to establish the EDI Holding Company to start the restructuring process. The process continues and will, no doubt, still face many obstacles as it seeks to cut through competing local government interests.

This lengthy narrative of the restructuring process in the distribution industry, indicates how local government can often frustrate rational proposals to create more efficient business entities which can deliver better services to customers. The complex web of local political interests, combined with a failure of leadership, could undermine the overall reform process and prevent potential benefits of lower cost electricity supply from reaching customers.

2002 – Eskom Corporatization

The first real evidence of ESI restructuring over the past decade has been the formal corporatization of Eskom, although the initiative came not from stakeholders involved in energy and electricity sector policy development, but from the Department of Public Enterprises (DPE) to which Eskom reports. Cabinet instructed the Department to give priority to the restructuring of the four largest state-owned enterprises: Eskom, Transnet, Telkom and Denel (an arms manufacturer).

A media release by the Minister of Public Enterprises, Jeff Radebe (dated 10 August 2000) describes the benefits of restructuring state-owned enterprises as follows:

“The economic growth and employment benefits of restructuring will be direct. It will result in increased investment, higher capital inflows and a reduction in public sector debt, with a significant longer term impact on South Africa’s GDP growth and substantial net proceeds accruing to the fiscus.”

“A Policy Framework: An Accelerated Agenda towards the Restructuring of State Owned Enterprises” was published by the Ministry in August 2000. The initial focus has been on clarifying the governance arrangements, including share-owner compacts with the Ministry, and obligations for the payment of taxes and dividends. The Minister of Public Enterprises, states in the foreword of the policy that:

“..government’s policy with regard to State Owned Enterprises is more properly referred to as a *restructuring* programme, and not in the more simplistic terms of privatization. The programme was and remains designed around a multiple array of strategies, or mixes of options, that are designed to ensure the maximization of shareholder interests defined in economic, social and development terms. Thus restructuring refers to the matrix of options that include the redesign of business management principles within enterprises, the attraction of strategic equity partnerships, the divestment of equity either in whole or in part where appropriate, and the employment of various immediate, turnaround initiatives.

“At the enterprise and sector level, restructuring involves improving the efficiency and effectiveness of the entity, accessing globally competitive technologies where appropriate, mobilizing private sector capital and expertise, and assisting the creation of effective market structures in sectors currently dominated by the SOEs. At a broader, macroeconomic level, restructuring initiatives aim to attract private direct investment, to contribute to the reduction in the public borrowing requirement, and to assist the development of an economic context that promotes industrial competitiveness and finances growth. Social imperatives include the need to ensure growth in employment, particularly in new areas of endeavor, and to rationalize or develop new skills within the labor force and their deployment throughout the country.”

The document states further that:

- “Eskom will be corporatized, with transmission, distribution and generation each forming a separate corporate entity.
- Different generating companies will be formed to promote internal competition prior to the introduction of private sector participation in generation, in conjunction with new power requirements.”

The report suggests that Transmission would probably remain in the hands of the state and that it is likely to take the form of a separate independent company.

The Eskom Conversion Act of 2001 replaced the old Eskom Act of 1987 and subsequent amendments. It converts Eskom into a public company (named Eskom Holdings Limited) with its share capital held by the state. This is the first step in restructuring Eskom which now pays taxes and dividends to the state.

A memorandum to the Bill described its purpose as bringing about more efficiency and competitiveness in the running of Eskom, exposing Eskom to global trends and ensuring that Eskom is run in terms of a protocol on cooperative governance. It further stated that “the benefit of the country of the unbundling of Eskom is that in the long run Eskom might decide to draw on the benefits of listing on a stock exchange and, in that event citizens and foreigners alike will be in a position to acquire shares in Eskom. Eskom can only be listed on a stock exchange if it is a company and Government is preparing for that eventuality, should it arise.”

There was strong opposition to this bill from organized labor. It argued that government had not followed the procedures agreed in the National Framework Agreement (NFA) whereby representatives of government and the union would negotiate the restructuring of individual SOEs. In May and June 2001, Cosatu made a submission on the Eskom Conversion Bill to the Public Enterprise Parliamentary Portfolio Committee. Its opposition centered on three main concerns: the Bill would pave the way for the privatization of Eskom; taxation of Eskom would impinge on its developmental role; and taxation would result on upward pressure on electricity prices. Agreement was reached on inserting new clauses into the Bill regarding the developmental role of Eskom and the protection of employees. However, they did not win the argument about Eskom paying taxed and dividends, and the issue of Eskom’s privatization remains open.⁵⁰

Labour has become increasingly alienated from government. Gwede Mantashe, the general-secretary of the National Union of Mineworkers, warned at a rally in Johannesburg that the ANC should not take the support of workers for granted. “It must listen to the working class and get their support, or it should listen to big capital and lose their support”.⁵¹ Cosatu embarked on a political strike on 30 and 31 August 2001 and marched to Parliament in protest against the government’s plans to privatize state assets.

⁵⁰ Tinto, E (2002). Restructuring South Africa’s Electricity Supply Industry. MPhil. University of Cape Town.

⁵¹ Business Day, 30 August 2001.

Ongoing – Unbundling, partial privatization and competition in the ESI

Since the publication of the Energy Policy White paper in 1998, momentum has been building around defining in more detail how the industry will be restructured to provide competition. This has occurred despite intense lobbying by Eskom to delay and limit the restructuring process.

In one of the rare occasions of World Bank involvement in South Africa, it sponsored a Ministerial Workshop on Electricity Supply Industry Reform held from 3-5 April 2000 in Midrand. The Minister of Minerals and Energy stated at the workshop that government's main objectives of reform are to:

- increase economic efficiency in investment decisions and operation so that costs and prices are as low as possible;
- maximise financial and economic returns to government from the ESI;
- increase the opportunity for black economic empowerment; and to
- protect public benefits such as widened access to the poor, energy efficiency ongoing R&D and environmental sustainability.

Different government departments seek different policy outcomes from electricity sector restructuring. The Minister of Minerals and Energy has consistently advocated that the industry should deliver electricity for all at affordable prices. She has also been a strong supporter of Black Economic Empowerment. The Minister of Public Enterprises has argued the benefits of restructuring SOEs to make them more efficient and to unlock economic value.⁵² The Ministry of Trade and Industry wishes to secure low prices to support the growth of a competitive economy, and National Treasury wishes to see growth in foreign direct investment, and would welcome additional fiscal receipts to reduce the national debt.

In summary, therefore, Government wishes to⁵³:

- *Keep future electricity prices as low as possible – but economically sustainable:* i.e. government wishes to ensure that investment (allocative) and operational efficiencies result in prices which are as low as possible – but prices are not so low so as to prejudice the financial viability of utilities.

Allocative efficiency: After many years of over-capacity, decisions have to be made soon around investments in the power generation plant. It is

⁵² Ministry of Public Enterprises (2000). An accelerated agenda towards the restructuring of state-owned enterprises: policy framework. August. Pretoria.

⁵³ Department of Minerals and Energy,(2000). Draft policy and strategy for electricity supply industry reform for the Republic of South Africa. Internal government paper. Version 5, April, Pretoria.

important that those decisions are made within the context of an industry structure that encourages efficient allocation of capital. Within the current industry structure, the costs of poor investment decisions are simply passed on to the consumer. Many argue that price increases⁵⁴ in the future could be held to a minimum when investors bear a more equitable share of the risk and hence are incentivized to make least-cost investments.

Productive efficiencies: International experience has demonstrated that the introduction of competition in generation has, in many cases, resulted in improved productivity and lower costs. If regulatory regimes ensure that these benefits also reach consumers (and not just investors) then lower prices will result.

- *Unlock economic value:* Government wishes to increase financial and economic returns from its state owned enterprises. To date it has received little financial return from the vast publicly owned resources that are invested in Eskom, nor has it received any taxation income from Eskom⁵⁵. Government can restructure the ESI in order to optimize the value of the industry and the financial returns it will receive, either in the form of annual cash flows or eventual lump sum proceeds from privatization. This income could be earmarked for more transparent funding of the electrification programme as well as funding other public objectives such as the reduction of state debt. There will be an important trade-off between maximizing returns from privatization and ensuring that electricity assets are not valued at too high a level that could result in high electricity prices for consumers because of the required revenue streams for investors to meet targeted rates of return.
- *Broaden economic ownership:* South Africa needs to widen the participation of the majority of its people in the economy. Eskom is one of the largest and best performing of the State's assets. Eskom generation assets provide extraordinary opportunities for promoting black economic empowerment through a discounted equity offering, with re-payment in dividends and earnings. An initial public offering (IPO) could enable broader public participation.
- *Increase foreign direct investment:* South Africa has inadequate domestic savings and needs to attract international capital into fixed domestic investment to support economic growth. International experience indicates that there is a great deal of interest in investing in existing and new

⁵⁴ Present average wholesale electricity prices are well below sustainable prices to pay for new generation investments.

⁵⁵ It should be noted, however, that – Eskom has, in the past, financed and cross-subsidized the huge national electrification programme.

electricity generation assets, particularly where the rules of the market are clear.

The World Bank sponsored seminar brought to South Africa a number of experts with detailed knowledge of the reform experience in their own countries. There was no single ideologically-inspired message or proposed model. Yet all advocated the merits of competition, but warned of the importance of careful design of the electricity market. At the end of the workshop senior government officials, including representatives from Eskom and the NER agreed to a draft policy paper on restructuring the ESI.

Eskom's top leadership, in the meantime, was alarmed at the extent of the reform proposals, particularly the recommendation to reduce Eskom's market share of generation to 35%. It lobbied at the very highest levels in government, drawing on its reputation for delivering low prices and for supporting government's RDP goals and its growing vision of an African renaissance, embodied in early versions of the New Partnership for African Development (NEPAD).

In May 2001, the Cabinet approved proposals for the reform of the ESI through a "managed liberalization" process. The elements of this are summarized below:⁵⁶

- *Structure of the generation industry:* Eskom is expected to retain no less than 70% of the existing electricity generation market, with privatisation of the remainder, with the initial aim of transferring 10% to black economic ownership no later than 2004;
- *Vertical unbundling:* to ensure non-discriminatory and open access to the transmission lines, a separate state owned Transmission Company will be established, independent of generation and retail businesses, with ring-fenced transmission system operation and market operation functions. Initially this transmission company would be a subsidiary of Eskom holdings and would be established as a separate state-owned transmission company before any investments are made in generation capacity;
- *Market structure:* Over time a multi-market model electricity market framework will ensure that transactions between electricity generators, traders and power purchasers may take place on a variety of platforms, including bilateral deals, power exchange and a balancing market. The market design should facilitate both physical and financial hedging; and
- *Regulation:* A regulatory framework will be put in place that ensures the participation of IPPs and the diversification of primary energy sources.

⁵⁶ Media briefing by Minister Phumzile Mlambo-Ngcuka, 3 July 2001, Pretoria.

In an agreement which originated at a summit at Farm Hill in October 2000, and which was signed on the 15 March 2002, the Department of Minerals and Energy (DME), the Department of Public Enterprises (DPE), the NER and Eskom reached broad consensus on the next steps in reform. An ESI restructuring committee, chaired by DPE, would be established and a detailed ESI Framework Paper would be completed in the course of 2002. Eskom would ring-fence its generation stations into clusters or portfolios for internal competition. Eskom Transmission would ring-fence its operations into wires, system operations, market operations and the international trader. The agreement further envisaged that Eskom Holdings would establish subsidiary companies for Eskom Generation and Eskom Transmission (although the latter issue was contested by Eskom). The internal pool would be converted into an independent market operation company (power exchange).

Although there have been general briefings to the Parliamentary Portfolio Committees, few details of the agreement have been made public. By the end of 2002, stakeholders had not been afforded an opportunity to publicly respond to government's reform proposals.

The DPE has established an ESI restructuring office and detailed studies have been undertaken on the clustering of Eskom generation plant and the creation of an electricity market, including a voluntary power exchange with a day-ahead-market, a balancing market, a market for ancillary services and range of other electricity trading platforms, including bilateral contracts and financial hedging instruments.

Organized labor (Cosatu) remains implacably opposed to any proposals to restructure the electricity industry. In 2002 they again embarked on a political national strike and protested against the possible privatization of Eskom and other utilities and the affects that this could have on the poor. The strike caused a prominent and acrimonious interchange between Cosatu and the government, with the latter insisting that it would not be deflected from its restructuring agenda.

The evolving agenda

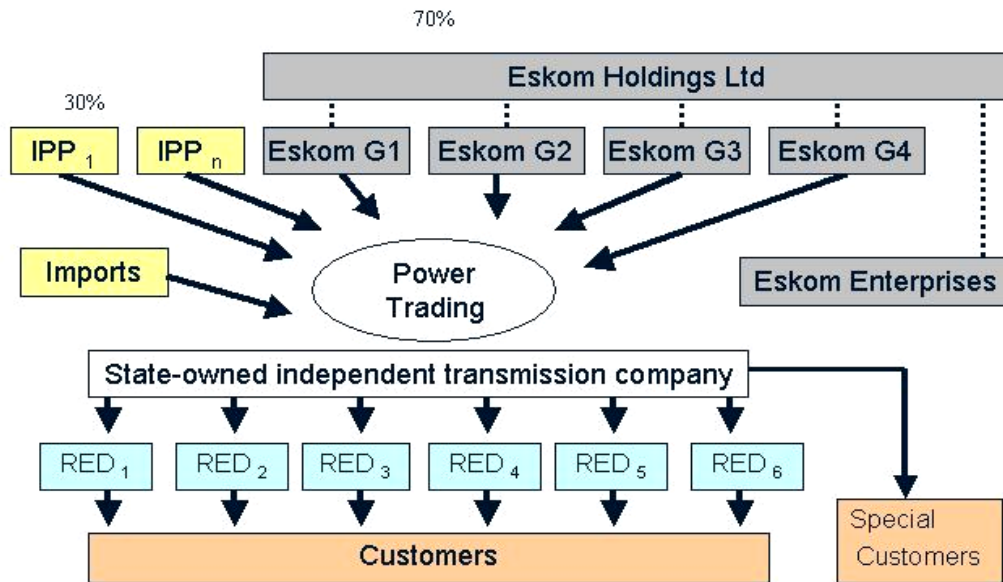
If the agreement reached between DPE, DME, NER and Eskom is finally approved by Cabinet, then the structure of the ESI in South Africa might look as follows by the year 2004:

An independent transmission company and power exchange with transparent market rules plus parallel trading mechanisms will give potential investors greater confidence. However, investors argue that there are still a number of potential problems with this model.

The positioning of generation clusters or subsidiaries under the Eskom Holdings Company could send the wrong message to the market. New entrants will not believe that the Eskom subsidiary companies are not acting in collusion. One way to ameliorate this situation would be to convert the generating clusters into separate subsidiary companies and to ensure that shareholder agreements, performance and management contracts make it

clear that each generation subsidiary should compete to maximize its own position. The existence of a Holdings company compromises this goal. The result could be that privatization receipts of those stations sold-off will be lower, and/or new entrants may be unwilling to participate in the market or may demand non-competitive long-term power purchase agreements (PPAs).

Figure 5: Likely ESI model in 2004



Eskom generation subsidiaries with open access and trading

Government's stated reason for reserving a dominant share (70%) of the generation market is not well understood. DME and DPE suggested to Cabinet that: "In order to meet Government's developmental objectives, Eskom will retain no less that 70% of the existing electricity generating market"⁵⁷. If the reference to "development" means electrification, then it does not make sense as Eskom will no longer be involved after the creation of the REDs. If it refers to affirmative procurement practices - these conditions could be included in any future privatization deal. If it refers to supporting the New Partnership for African Development (NEPAD) - Eskom Enterprises is Eskom's vehicle for going into Africa, not Eskom Generation in South Africa. Investors argue that there is no logic to this policy and that Eskom's share of the generation market could and should be reduced to below 35%. This proposal accepts that competition requires a dilution of Eskom's market

⁵⁷ Media briefing by Minister Phumzile Mlambo-Ngcuka, 3 July 2001, Pretoria

power and significant entry of new players. Divested assets could be privatized through black economic empowerment provisions, an IPO or private equity participation.

Proponents of this model argue that competition is likely to result in improved efficiency and lower prices than the previous models. Significant black economic empowerment could be achieved as well as fiscal revenue for debt reduction. Significant inward investment could also result. Eskom would be forced to compete, and a management culture and practice might be established which could position Eskom better to invest abroad and contribute to NEPAD. Southern African regional economic development could be significantly enhanced as generation projects in the region could have access to the South African market and would operate with greater market certainty.

In addition, the fact that the state retains ownership of a portion of generation would enable it to intervene more easily in new generation investments should this be necessary if the market fails to respond to growing demand.

This model gives practical expression to government policy as laid out in the Energy Policy White Paper of 1998. The Energy White Paper committed government to “introduce competition to the industry, especially the generation sector”, “permit open, non discriminatory access to the transmission system” and “encourage private sector participation in the industry”.

V. Conclusion

There are elements of power sector reform in South Africa which are peculiar to its recent history, namely its transformation into a democratic state after the long, dark years of apartheid repression. Within this context, it was inevitable that energy policy would be transformed from a defensive obsession with security to a new focus on promoting social equity and improving economic competitiveness as South Africa re-integrated with the global economy. The Energy Policy White paper gave expression to this policy shift, but it was already evident in the launch of an impressive electrification programme that sought to tackle the huge backlog of the previously disenfranchised’s demand for affordable access to electricity. Yet this first significant change in the ESI after the political transformation of the country was not linked in any obvious way to the restructuring of the electricity industry. The state-owned utility Eskom, in partnership with local government distributors, were able to mount an accelerated electrification programme without restructuring. Surplus and cheap electricity was available as a result of over-investment in the previous decade, and a strong, large industrial consumer base enabled the ESI to cross-subsidize the electrification programme without the necessity of imposing unaffordable price hikes.

The second broad area for action and change in the 1990s, flowed from the above concern to ensure sustainable delivery of affordable electricity to the majority of the population. Consensus soon developed amongst political parties, government, trade unions and civil society groupings that the fragmentation and inefficiency of the distribution side of the ESI would need to be overhauled. The distribution business of Eskom would need to be combined with the hundreds of local government distributors and rationalized into a limited number (6) of regional electricity distributors. The process has been slow and has been frustrated by the complex web of political interests at local government level and the fear of loss of control of an important infrastructure service and large income streams. Nevertheless, the process of reform in this sector continues and government is intent on creating a more efficient industry in the form of new, commercially-run, public corporations.

This emphasis on corporatization reflects a general commitment to re-assess government's role in the economy, particularly the SOEs in the infrastructure sector. In the 1980s, government began to examine the governance of these enterprises and the way they were funded and operated financially. Eskom's governance was reformed, government guarantees were withdrawn and its capital expenditure programme and operations were placed on a more commercial footing. This process of reform was strengthened after 1994, when the new democratic government placed even more emphasis on conservative fiscal management and the liberalization of the economy. Eskom, along with other SOEs, was corporatized, had to pay taxes and dividends, and was subject to a shareholder performance contract. At the same time, the relationship of the state to the sector was clarified through the creation of an independent electricity regulator which approves prices without political interference.

Although the liberalization and restructuring of the ESI in South Africa is far behind that in many other countries, the reform process continues, informed, in general, by government's commitment to increase the competitiveness of the economy and also to broaden economic participation and ownership for black South Africans. The momentum for reform has been set by the broad direction of economic policy. It was also been set in motion through the work of analysts that brought international experience of power sector reform to bear in the Energy Policy White Paper, and who argue that South Africa is living on borrowed time in terms of low electricity prices. Arguments are now being made that a vertically integrated, state-owned, monopoly industry, even if it is corporatized, is unlikely to make efficient investment decisions. The current low electricity prices are primarily a result of investment contraction after a previous period of wasteful over-investment. The key challenge for the ESI is to create a competitive structure where investors bear a more equitable share of the risk, thus creating an environment for more efficient allocative and operational efficiency. Government has made a broad commitment to manage the liberalization of the ESI and the introduction of competition. Detailed planning has begun.

However, government still experiences ambivalence and doubts around embarking on a path of full unbundling, competition and privatization. Eskom is still seen as an important instrument of government policy, a well-performing infrastructural industry that supports government's economic and social programme. Current low prices create a false complacency. And government faces serious resistance from organized labor, who has picked issues around Eskom reform as the battle-ground against government's commitment to privatization. In the next years, it could be the interests of the new black economic elite, interested in a share of privatization receipts that sustains the momentum for reform. Ultimately, the overall context of economic liberalization (managed and regulated where appropriate) will sharpen the imperative of being competitive and efficient, and the reform of the electricity industry in South Africa will continue.